

# Alessandro Casa

ASSOCIATE PROFESSOR · FACULTY OF ECONOMICS AND MANAGEMENT · FREE UNIVERSITY OF BOZEN-BOLZANO

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## Research Interests

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**Keywords:** Latent variable models, Nonparametric statistics, Density-based clustering, Density estimation, Computational statistics, Dimensionality reduction techniques, Graphical models, Bayesian Inference, Penalized estimation, Chemometrics.

## Professional Experience

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### ➤ CURRENT POSITION

Oct 2024 - **Associate Professor of Statistics**, (STAT-01/A)  
Faculty of Economics and Management, Free University of Bozen-Bolzano, Italy

### ➤ PAST POSITIONS

Feb 2024 - **Senior Assistant Professor**, (RTDB - SECS-S/01)  
Sep 2024 Department of Economics, University of Bergamo, Italy  
Jan 2022 - **Junior Assistant Professor**, (RTDA - SECS-S/01)  
Jan 2024 Faculty of Economics and Management, Free University of Bozen-Bolzano, Italy  
Mar 2020 - **Postdoctoral Research Fellow**, (supervisor: T.B. Murphy)  
Dec 2021 School of Mathematics and Statistics, University College Dublin, Ireland  
Affiliated with: Vistamilk SFI Research Centre, Insight SFI Research Centre  
Nov 2019 - **Research Assistant Pending Viva**, (supervisor: T.B. Murphy)  
Feb 2020 School of Mathematics and Statistics, University College Dublin, Ireland  
Affiliated with: Vistamilk SFI Research Centre, Insight SFI Research Centre

## Awards & Scholarships

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2023 **National Qualification for Associate Professor**, (ASN II fascia, Statistics, SC 13/D1)  
2023 **PhD Student/Young Researcher Paper Award for outstanding contribution to the field of Classification, Clustering and Data Analysis**, CLADAG - Classification and Data Analysis Group  
2022 **Best contribution by a young researcher**, Italian Statistical Society (14th Scientific Meeting)  
2016 **Doctoral Scholarship**, University of Padova

## Education

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### University of Padova

Padova (Italy)

#### PHD IN STATISTICS

Oct 2016 - Sep 2019

- Thesis: Climbing modes and exploring mixtures: a journey in density-based clustering
- Advisor: Prof. G. Menardi, Co-Advisor: Prof. J.E. Chacón

### University of Padova

Padova (Italy)

#### MASTER DEGREE IN STATISTICAL SCIENCES

Oct 2014 - Sep 2016

- Thesis: Semi-supervised detection of collective anomalies using non-parametric methods with an application to particle physics (in italian)
- Grade: 110/110 cum laude
- Advisor: Prof. G. Menardi

## University of Padova

### BACHELOR DEGREE IN STATISTICS, ECONOMICS AND FINANCE

- Thesis: Dimension reduction for model-based clustering: a simulation study and an application to genetic data (in italian)
- Grade: 110/110 cum laude
- Advisor: Prof. G. Menardi

Padova (Italy)

Oct 2011 - Jul 2014

## Visiting periods

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### University of Cote d'Azur, Laboratory J.A. Dieudonné

#### VISITING PHD SCHOLAR

- Advisor: Prof. C. Bouveyron

Nice (France)

Oct 2018 - Dec 2018

### University of Perugia, Department of Economics

#### VISITING PHD SCHOLAR

- Advisor: Prof. L. Scrucca

Perugia (Italy)

Apr 2018

### University of Cambridge, Department of Pure Mathematics and Mathematical Sciences

#### VISITING PHD SCHOLAR

- Advisor: Prof. R. Samworth

Cambridge (UK)

Oct 2017 - Dec 2017

## Presentations

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### ➤ SEMINARS

May 2025 | Research Seminar Series, Institute of Statistics and Mathematics. Wirtschaftsuniversität Wien

May 2025 | Seminar Series, Department of Statistical Sciences. Università Cattolica del Sacro Cuore

Mar 2025 | DSS Statistics Seminar. Sapienza Università di Roma

Mar 2024 | Brown Bag Seminar. University of Bergamo

May 2023 | Mathematics for Data Science, AI and ML Seminar Series. University of Trento

Feb 2023 | Working Group on Statistical Learning. University College Dublin

Feb 2022 | Statistics Seminar Series. Trinity College Dublin

Feb 2021 | Working Group on Statistical Learning. University College Dublin

Jul 2020 | Vistamilk Spectroscopy Research Meeting. Online

Feb 2020 | Working Group on Statistical Learning. University College Dublin

### ➤ INVITED TALKS

Dec 2025 | [CMStat25] 19th International Conference on Computational and Methodological Statistics. London

Jun 2025 | [SIS25] Intermediate conference of the Italian Statistical Society. Genova

Dec 2024 | [CMStat24] 18th International Conference on Computational and Methodological Statistics. London/Online

Jul 2024 | [WG-MBC24] Working Group on Model-Based Clustering. Bertinoro (FC)

Jun 2024 | [SIS24] 52nd Scientific Meeting of the Italian Statistical Society. Bari

Sep 2023 | [CLADAG23] SIS Classification and Data Analysis Group. Salerno

Jul 2023 | [WG-MBC23] Working Group on Model-Based Clustering. Pittsburgh

Dec 2022 | Workshop on Statistical Learning and Econometrics. Bolzano

Dec 2022 | [CMStat22] 15th International Conference of the ERCIM Working Group. London/Online

Sep 2022 | Vistamilk Science Foundation Ireland Site Review. Online

Sep 2022 | [ECDA22] European Conference on Data Analysis. Naples  
 Aug 2022 | [COMPSTAT22] 24th International Conference on Computational Statistics. Bologna  
 Jun 2022 | [SIS22] 51st Scientific Meeting of the Italian Statistical Society. Caserta  
 Dec 2021 | [CMStat21] 14th International Conference of the ERCIM Working Group. Online  
 Mar 2021 | Vistamilk Annual Conference. Online  
 Oct 2020 | Vistamilk Science Foundation Ireland Site Review. Online  
 Sep 2019 | [CLADAG19] SIS Classification and Data Analysis Group. Cassino (FR)  
 Jul 2019 | [EMS19] European Meeting of Statisticians. Palermo

#### ➤ OTHER TALKS AND POSTER

Jul 2025 | [IWSM25] International Workshop on Statistical Modelling. Limerick  
 Oct 2021 | [WG-MBC21] Working Group on Model-Based Clustering. Athens  
 Sep 2019 | [CLADAG21] SIS Classification and Data Analysis Group. Online  
 Jun 2021 | [SIS21] 50th Scientific Meeting of the Italian Statistical Society. Online  
 Mar 2021 | [CASI21] 41st Conference on Applied Statistics in Ireland. Online  
 Jul 2019 | [WG-MBC19] Working Group on Model-Based Clustering. Wien  
 Sep 2018 | [ASPD] Workshop on Advanced Statistics for Physics Discovery. Padova  
 Sep 2018 | [MBC2] Workshop on Model-Based Clustering and Classification. Catania  
 Jul 2018 | [ECDA18] European Conference on Data Analysis. Paderborn  
 Jun 2018 | [SIS18] 49th Scientific Meeting of the Italian Statistical Society. Palermo  
 Jun 2018 | [ISNPS] International Society for Nonparametric Statistics. Salerno  
 Jun 2017 | [SIS17] Intermediate conference of the Italian Statistical Society. Florence

#### ➤ CHAIR & DISCUSSANT

Dec 2025 | [CMStat25] 19th International Conference on Computational and Methodological Statistics  
 Organizer and chair of the session “Recent development in Bayesian clustering”. London  
 Jun 2024 | [SIS24] 52nd Scientific Meeting of the Italian Statistical Society  
 Chair of the session “Advanced statistical methods to analyze complex data”. Bari  
 2020 - 2024 | Chair of several sessions at the Vistamilk Science Foundation Ireland Site Review  
 and at the International Workshop on Spectroscopy and Chemometrics

## Students Supervision

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Piero Caruso (2025, MSc in Data Analytics for Economics and Management, unibz, co-supervised with prof. D. Ferrari),  
 Alice Guglielmino (2025, BSc in Economics and Management, unibz), Elisa Borrini (2024, MSc in Mathematical Engineering,  
 Politecnico di Milano, co-supervised with prof. A. Cappozzo), Jacopo Gardella (2024, MSc in Mathematical Engineering,  
 Politecnico di Milano, co-supervised with prof. A. Pini and prof. R. Argiento), Isabel Franchini (2023, MSc in Entrepreneurship  
 and Innovation, unibz), David Bottai (2023, MSc in Entrepreneurship and Innovation, unibz, co-supervised with prof. A.  
 Narduzzo), Giulia Pascali (2019, MSc in Statistical Sciences, unipd, co-supervised with prof. G. Menardi)

## Teaching Activities

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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| from 24/25 | <b>Statistical Methods for Business Analysis (english)</b> , Lecturer, MSc in Data Analytics for Economics and Management, MSc in Entrepreneurship and Innovation (36hrs) | unibz |
| - present  |                                                                                                                                                                           |       |
| from 24/25 | <b>Research Methods and Experimental Design (italian 24/25, english 25/26)</b> , Lecturer, MSc in Tourism Management (36hrs)                                              | unibz |
| - present  |                                                                                                                                                                           |       |
| from 24/25 | <b>Statistics for Tourism, Sport and Event Management (italian)</b> , Lecturer, BSc in Tourism Sport and Event Management (54hrs)                                         | unibz |
| - present  |                                                                                                                                                                           |       |
| a.y. 23/24 | <b>Statistics (italian)</b> , Lecturer BSc in Mechanical Engineering (48hrs)                                                                                              | unibg |
| a.y. 22/23 | <b>High-dimensional Econometric Methods for Big Data (english)</b> , Lecturer PhD in Economics and Finance (8hrs)                                                         | unibz |
| from 22/23 | <b>Statistics for Tourism, Sport and Event Management (italian)</b> , Teaching Assistant, BSc in Tourism Sport and Event Management (18hrs)                               | unibz |
| - to 23/24 |                                                                                                                                                                           |       |
| a.y. 22/23 | <b>Statistics (italian)</b> , Teaching Assistant BSc in Economics and Management (18hrs)                                                                                  | unibz |
| from 21/22 | <b>Statistical Methods for Business Analysis (english)</b> , Lecturer, MSc in Entrepreneurship and Innovation (52hrs, 54hrs)                                              | unibz |
| - to 22/23 |                                                                                                                                                                           |       |
| a.y. 19/20 | <b>STAT20060 - Statistics &amp; Probability</b> , Lecturer, Level 2 (9hrs)                                                                                                | ucd   |
| a.y. 19/20 | <b>STAT20060 - Statistics &amp; Probability</b> , Teaching Assistant, Level 2                                                                                             | ucd   |
| a.y. 19/20 | <b>STAT30270 - Statistical Machine Learning</b> , Teaching Assistant, Level 3                                                                                             | ucd   |
| a.y. 18/19 | <b>Statistica Iterazione (italian)</b> , Specialist lecture “Bike sharing in Paris: a case study”, MSc in Statistical Sciences                                            | unipd |

## Funding and Grants

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|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan 2026 -  | <b>Co-Principal Investigator</b> , of the unibz unit. “Learning complex networks for economic data: theory, methods, and algorithms”, Swiss National Science Foundation grant (≈€600k)                                                                  |
| Dec 2028    |                                                                                                                                                                                                                                                         |
| Feb 2025 -  | <b>Principal Investigator</b> , “SPEX - Consumer behavior, digital platforms and spatial spillover effects in a dual electricity market”, Progetto PRIN 2022 PNRR, cod. progetto P20227RKF9 (CUP: I53D23006190001)                                      |
| Feb 2026    |                                                                                                                                                                                                                                                         |
| Jan 2025 -  | <b>Principal Investigator</b> , “SIERRA - Advanced Statistical Methods for Spectral Data”, Start-up fund, Free University of Bozen-Bolzano (€50k)                                                                                                       |
| Dec 2027    |                                                                                                                                                                                                                                                         |
| Jan 2022 -  | <b>Member of the research group</b> , “Analisi di dati ad alta dimensionalità per processi decisionali sostenibili”, (CUP I51B21004310005), scientific supervisor: Prof. D. Ferrari                                                                     |
| Jan 2024    |                                                                                                                                                                                                                                                         |
| Nov 2019 -  | <b>Member of the research group</b> , “Insight Science Foundation Ireland Research Centre for Data Analytics” (ref. no. SFI/12/RC/2289_P2), scientific supervisor: Prof. T.B. Murphy                                                                    |
| Dec 2021    |                                                                                                                                                                                                                                                         |
| Nov 2019 -  | <b>Member of the research group</b> , Vistamilk - Research grant from Science Foundation Ireland and the Department of Agriculture, Food and Marine on behalf of the Government of Ireland (grant 16/RC/3835), scientific supervisor: Prof. T.B. Murphy |
| Dec 2021    |                                                                                                                                                                                                                                                         |
| 2019        | <b>Member of the research group</b> , “Approfondimenti su metodi di clustering basati sulla densità”, scientific Research Project DOR (prot. DOR1944143), scientific supervisor: Prof. G. Menardi                                                       |
| 2018        | <b>Member of the research group</b> , “Bandwidth selection in nonparametric density estimation for clustering purposes”, scientific Research Project DOR (prot. DOR1898215), scientific supervisor: Prof. G. Menardi                                    |
| 2016 - 2019 | <b>Member of the research group</b> , “AMVA4NP - Advanced Multivariate Analysis for New Physics Searches at the Large Hadron Collider”, Marie Skłodowska Curie - Innovative Training Network (Horizon 2020), scientific supervisor: Prof. G. Menardi    |

## Service & Outreach

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### ➤ SERVICE

- 2025 - 2027 **Scientific Board Member**, Classification and Data Analysis Group (CLADAG) of the Italian Statistical Society
- Mar 2025 - **Quality Assurance responsible**, Faculty of Economics and Management, Free University of Bozen-Bolzano
- present 2025 - **Member of the Admission Board**, MSc in Data Analytics for Economics and Management and BSc in Tourism, Sport and Event Management, Free University of Bozen-Bolzano
- Jan 2022 - **Member of the Selection committee**, Evaluation and ranking of the applications for
- Jan 2024 Teaching contract, Lecturing assistant and Teaching assistant positions, Free University of Bozen-Bolzano
- 2022 - 2023 **Member of the degree council**, MSc in Entrepreneurship and Innovation, Free University of Bozen-Bolzano
- 2016 - 2020 **Elected representative of PhD Students and Postdoctoral researcher**, Faculty Board, University of Padova
- 2016 - 2019 **Elected representative of PhD Students**, Doctoral Board of the Grad School, University of Padova

### ➤ OUTREACH

- Jun 2025 **Organizer**, “unibz Data4Econ Hackathon”, first edition of a student hackathon dedicated to data analytics for economics. Sponsors: Alperia, Gruppo Dolomiti Energia
- 2022 & **Organising committee of the Math & Stat area**, “Le mille e una scienza”, youth-oriented 2023 science festival in Bolzano
- 2020 **Participant**, “I’m a scientist - Get me out of here”, event in Ireland connecting school students with scientists
- 2016 & 2019 **Participant**, “Notte europea dei ricercatori”, University of Padova

### ➤ ORGANIZATION OF SCIENTIFIC EVENTS

- Sep 2026 **Member of the scientific committee**, European Conference on Data Analysis (ECDA), (Stralsund, Germany)
- Jul 2026 **Member of the local organising committee**, 19th Conference of the International Federation Of Classification Societies (IFCS), (Milan, Italy)
- Nov 2025 **Member of the organising committee**, PRIN Workshop “Spatial Econometrics meets Energy Economics: A workshop on methods, data and applications”, (Bolzano, Italy)
- Apr 2024 **Member of the organising committee**, 4th edition of the “International Workshop on Spectroscopy and Chemometrics”, (Fermoy, Ireland)
- Dec 2022 **Member of the organising committee**, “Workshop on statistical learning and econometrics - Methodological advancements and applications”, (Bolzano)
- Apr 2022 **Member of the organising committee**, 2nd edition of the “International Workshop on Spectroscopy and Chemometrics”, (Dublin, Ireland)
- Apr 2021 **Academic lead of the organising committee**, 1st edition of the “International Workshop on Spectroscopy and Chemometrics”, (online)
- Jul 2020 - **Member of the organising committee**, Monthly Vistamilk Spectroscopy Research Meeting Nov 2024

## ➤ EVENTS PARTICIPATION

- Jun 2018 **Start Up Research - Follow up**, Follow up meeting where research groups presented and discussed the scientific output of their works, (Palermo)
- Jun 2017 **Start Up Research**, Meeting where small research groups of young scholars, advised by senior researchers, were asked to develop innovative methods to analyze a common dataset, (Siena)
- Jun 2017 **Stats Under the Stars 3**, Hackathon, organizers: University of Florence, Italian Statistical Society (Florence)
- Sep 2015 **Stats Under the Stars 1**, Hackathon, organizers: University of Padova, Italian Statistical Society (Florence)
- Jun 2015 **Data Science Game**, Hackathon, organizers: Google, Capgemini (Paris)

## ➤ EDITORIAL ACTIVITY

**Referee for:** Advances in Data Analysis and Classification, Annals of Applied Statistics, Bayesian Analysis, Canadian Journal of Statistics, Computational Statistics and Data Analysis, Electronic Journal of Statistics, Food Analytical Methods, Journal of Applied Econometrics, Journal of Classification, Journal of Computational and Graphical Statistics, Journal of Dairy Science, Journal of Royal Statistical Society: Series B, Journal of Royal Statistical Society: Series C, Journal of Statistical Software, Knowledge-based Systems, Metron, New England Journal of Statistics in Data Science, R Journal, Scientific Reports, Social Indicators Research, Stat, Statistical Analysis and Data Mining, Statistics and Computing, Statistics and Probability Letters, Statistics in Medicine, WIREs Computational Statistics.

## ➤ MEMBERSHIPS

- 2021 - **Member of the Italian Statistical Society (SIS)**, Affiliated with Classification and Data Analysis Group (CLADAG), Research Group for Statistical Applications to Environmental Problem (GRASPA), young SIS (ySIS)
- 2021 - **Member of the Irish Statistical Association (ISA)**, Affiliated with the young ISA (yISA)

## ➤ LANGUAGES

**Italian:** mother tongue

**English:** C1, Certificate of the Language center of the Free University of Bozen-Bolzano

**German:** A2, Goethe zertifikat

## Publications

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### ➤ ARTICLES UNDER REVIEW

- Casa, A.**, Murphy, T.B. & Fop, M. (2025+). A latent variable model for identifying and characterizing food adulteration. *arXiv preprint arXiv:2512.13939*
- Gardella, J., Argiento, R., **Casa, A.** & Pini, A. (2025+). Addressing Phase Discrepancies in Functional Data: A Bayesian Approach for Accurate Alignment and Smoothing. *arXiv preprint arXiv:2506.14650*

### ➤ ARTICLES IN REFEREED JOURNALS (STATISTICS)

- Casa, A.**, Ferrari, D. & Huang, Z. (2025). High-dimensional covariance estimation by pairwise likelihood truncation. *Biometrika*, just accepted, <https://doi.org/10.1093/biomet/asaf087>
- Casa, A.** & Ferrari, D. (2025). Confidence set for mixture order selection. *Statistics and Probability Letters*, 226, 110509
- Cappozzo, A. & **Casa, A.** (2025). Model-based clustering for covariance matrices via penalized Wishart mixture models. *Computational Statistics and Data Analysis*, 212, 108232
- Cappozzo, A., **Casa, A.** & Fop, M. (2025). Sparse model-based clustering of three-way data via lasso-type penalties. *Journal of Computational and Graphical Statistics*, 34 (3), 1030-1050

- Ferraccioli, F., **Casa, A.** & Stefanucci, M. (2024). An adaptive functional regression framework for spatially heterogeneous signals in spectroscopy. *Journal of the Royal Statistical Society: Series C*, 73(5), 1370-1388.
- Casa, A.**, Cappozzo, A. & Fop, M. (2022). Group-wise shrinkage estimation in penalized model-based clustering. *Journal of Classification*, 39(3), 648-674.
- Casa, A.**, O’Callaghan, T.F. & Murphy, T.B. (2022). Parsimonious Bayesian Factor Analysis for modelling latent structure in spectroscopy data. *Annals of Applied Statistics*, 16(4), 2417-2436
- Casa, A.** & Menardi, G. (2021). Nonparametric semi-supervised classification with application to signal detection in high energy physics. *Statistical Methods & Applications*, 31(3), 531-550
- Casa, A.**, Bouveyron, C., Eroshova, E. & Menardi, G. (2021). Co-clustering of time-dependent data via the Shape Invariant Model. *Journal of Classification*, 38(3), 626-649
- Casa, A.**, Scrucca, L. & Menardi, G. (2021). Better than the best? Answers via model ensemble in density-based clustering. *Advances in Data Analysis and Classification*, 15(3), 599-623
- Casa, A.**, Chacón, J.E. & Menardi, G. (2020). Modal clustering asymptotics with applications to bandwidth selection. *Electronic Journal of Statistics*, 14(1), 835-856

## ➤ ARTICLES IN REFEREED JOURNALS (CROSS-DISCIPLINARY)

- Frizzarin, M., Hayes, E., **Casa, A.** & Berry, D.P. (2025). Comparison of the mid-infrared spectra and prediction equations developed from morning and evening milk samples from twice-a-day milked dairy cows. *Journal of Dairy Science*, 108(2), 1573-1583
- Frizzarin, M., Visentin, G., Ferragina, A., Hayes, E., Bevilacqua, A., Dhariyal, B., Domijan, K., Khan, H., Ifrim, G., Le Nguyen, T., Meagher, J., Menchetti, L., Singh, A., Whoriskey, S., Williamson, R., Zappaterra, M. & **Casa, A.** (2023). Classification of cow diet based on milk mid infrared spectra: a data analysis competition at the “International workshop on spectroscopy and chemometrics 2022”. *Chemometrics and Intelligent Laboratory Systems*, 104755.
- Frizzarin, M., Gormley, I.C., **Casa, A.** & McParland, S. (2021). Selecting milk spectra to develop equations to predict milk technological traits. *Foods*, 10(12), 3084
- Stakia, A., Dorigo, T., Banelli, G., Bortoletto, D., **Casa, A.**, de Castro, P., Delaere, C., Donini, J., Finos, L., Gallinaro, M., Giannamanco, A., Held, A., Jimenez Morales, F., Kotkowski, G., Liew, S.P., Maltoni, F., Menardi, G., Papavergou, I., Saggio, A., Scarpa, B., Strong, G.C., Tosci, C., Varela, J., Vischia, P. & Weiler, A. (2021). Advanced multi-variate analysis methods for new physics searches at the Large Hadron Collider. *Reviews in Physics*, 7, 100063.
- Frizzarin, M., Bevilacqua, A., Dhariyal, B., Domijan, K., Ferraccioli, F., Hayes, E., Ifrim, G., Konkolewska, A., Le Nguyen, T., Mbaka, U., Ranzato, G., Singh, A., Stefanucci, M. & **Casa, A.** (2021). Mid infrared spectroscopy and milk quality traits: a data analysis competition at the “International Workshop on Spectroscopy and Chemometrics 2021”. *Chemometrics and Intelligent Laboratory Systems*, 219, 10442
- Frizzarin, M., O’Callaghan, T.F., Murphy, T.B., Hennessy, D. & **Casa, A.** (2021). Application of machine learning methods to milk mid infrared spectra for discrimination of cows milk from pasture or TMR diets. *Journal of Dairy Science*, 104(12), 12394-12402
- Frizzarin, M., Gormley, I.C., Berry, D.P., Murphy, T.B., **Casa, A.**, Lynch, A. & McParland, S. (2021). Predicting cow milk quality traits from routinely available milk spectra using statistical machine learning methods. *Journal of Dairy Science*, 104(7), 7438-7447

## ➤ REFEREED DISCUSSIONS

- Casa, A.**, Fop, M. & D’Angelo, S. (2024). Contributed discussion on: “Sparse Bayesian factor analysis when the number of factors is unknown” by Frühwirth-Schnatter, S., Hosszejni, D. & Freitas Lopes, H. *Bayesian Analysis*, 20(1), 213-344
- Casa, A.**, Fop, M. & Murphy, T.B. (2021). Contributed discussion on: “Centered partition processes; informative priors for clustering” by Paganin, S., Herring, A.H., Olshan, A.F. & Dunson, D.B. *Bayesian Analysis*, 16(1), 301-370

## ➤ BOOK CHAPTERS

- Casa, A.**, Cappozzo, A. & Fop, M. (2023). Penalized Model-Based Clustering with Group-Dependent Shrinkage Estimation. In: *Building Bridges between Soft and Statistical Methodologies for Data Science. SMPS 2022. Advances in Intelligent Systems and Computing*, vol 1433. Springer. [https://doi.org/10.1007/978-3-031-15509-3\\_10](https://doi.org/10.1007/978-3-031-15509-3_10)

Cabassi, A., **Casa, A.**, Farcomeni, A., Fontana, M., Russo, M. (2018). Three testing perspectives on connectome data. *Studies in Neural Data Science, Springer Volume 'Proceedings in Mathematics & Statistics'*, ISBN-9783030000394.

## ➤ PH.D THESIS

**Casa, A.** (2020). Climbing modes and exploring mixtures: a journey in density-based clustering. PhD Thesis, University of Padova, <https://hdl.handle.net/10863/43031>

## ➤ REFEREED CONFERENCE PROCEEDINGS

**Casa, A.** & Ferrari, D. (2025). Model selection confidence sets for the number of mixture components. *Proceedings of the 39th International Workshop on Statistical Modelling* ISBN: 978-1-0369-2711-0

**Casa, A.** & Cappozzo, A. (2025). Penalized model-based clustering for covariance matrices. *Statistics for Innovation I - SIS 2025, Short Papers, Plenary, Specialized, and Solicited Sessions* ISBN: 978-3-031-96735-1

Gardella, J., **Casa, A.**, Argiento, R. & Pini, A. (2025). Bayesian blended landmark model for alignment of functional data. *Statistics for Innovation III - SIS 2025, Short Papers, Contributed Sessions 2* ISBN: 978-3-031-95994-3

**Casa, A.**, Murphy, T.B. & Fop, M. (2024). Sparse partial membership models with applications in food science. *Methodological and Applied Statistics and Demography I - SIS 2024, Short Papers, Plenary and Specialized Sessions* ISBN: 978-3-031-64345-3

Gardella, J., Carone, R., Ciarulli, S., Francescon, N., Freddi, M., Garlanda, E., **Casa, A.** & Pini, A. (2024). Bayesian models for registration of functional data. *Methodological and Applied Statistics and Demography III - SIS 2024, Short Papers, Contributed Sessions 1* ISBN: 978-3-031-64430-6

**Casa, A.**, Murphy, T.B. & Fop, M. (2023). Partial membership models for high-dimensional spectroscopy data. *CLADAG 2023 - Book of Abstracts and Short Papers*. ISBN: 9788891935632

Ferraccioli, F., **Casa, A.** & Stefanucci, M. (2022). Trend filtering for functional regression. *Book of the Short Papers, Pearson, Caserta, 22.6.2022 - 24.6.2022*, ISBN: 9788891932310

**Casa, A.**, Cappozzo, A. & Fop, M. (2022). Group-wise penalized estimation schemes in model-based clustering. *Book of the Short Papers. Pearson, Caserta, 22.6.2022 - 24.6.2022*, ISBN: 9788891932310

Cappozzo, A., **Casa, A.** & Fop, M. (2021). Model-based clustering with sparse matrix mixture models. *CLADAG 2021 - Book of abstract and short papers*. DOI: 10.36253/978-88-5518-340-6

Cappozzo, A., **Casa, A.** & Fop, M. (2021). Penalized model-based clustering for three-way data structures. *Book of short papers - SIS 2021*. ISBN 9788891927361

**Casa, A.**, O'Callaghan, T.F. & Murphy, T.B. (2021). Parsimonious modelling of spectroscopy data via a Bayesian latent variables approach. *Book of short papers - SIS 2021*. ISBN 9788891927361

**Casa, A.**, Chacòn, J.E. & Menardi, G. (2019). Asymptotics for bandwidth selection in nonparametric clustering. *Cladag 2019 - Book of short papers*. ISBN 9788883171086

Pascali, G., **Casa, A.** & Menardi, G. (2019). Co-clustering TripAdvisor data for personalized recommendations. *Book of short Papers SIS 2019*. ISBN 978889191510.

**Casa, A.**, Chacòn, J.E. & Menardi, G. (2018). On the choice of an appropriate bandwidth for modal clustering. *Book of short Papers SIS 2018*. ISBN 9788891910233

**Casa, A.** & Menardi, G. (2017). Signal detection in high energy physics via a semisupervised nonparametric approach. *Proceedings of the Conference of the Italian Statistical Society "Statistics and Data Sciences: new challenges, new generations"*. ISBN 9788864535210